

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 9 — Overlapping Generations

All 13 end-of-chapter exercises, 9.1–9.13 (pp. 366–378)

What is in this file

All thirteen exercises are analytical. They are pure-exchange Samuelson economies with fiat currency (9.1, 9.2, 9.4, 9.5), a Lucas tree (9.3), several currencies (9.6), borrowers and lenders inside a generation (9.7, 9.8, 9.10), social security (9.9), and money- or bond-financed deficits (9.10–9.13). One method runs through all of them. Write real balances as the saving of the young, turn the money market into a difference equation, and read off the stationary point, the continuum around it, and the Pareto ranking.

The companion notebook `ls_ch09.ipynb` (same folder) checks every key result numerically by an independent route with an `assert`. Saving functions are checked by numerical optimisation, closed-form price paths against the forward recursion, roots by `brentq` against the quadratic formula, and the Pareto claims by direct utility comparison. Where the book gives no numbers, the parameters are chosen and stated.

Question boxes reproduce the full statement. Blue boxes are odd-numbered exercises, purple boxes even-numbered ones.

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Chapter 9 Overlapping Generations

Chapter 9 Overview

Sequential trading with fiat money (section 9.4). A young agent born at i faces $c_i^i + m_i^i/p_i \leq y_i^i$ and $c_{i+1}^i \leq m_i^i/p_{i+1} + y_{i+1}^i$ with $m_i^i \geq 0$ (9.4.1)–(9.4.3). The gross real return on currency is $R_i = p_i/p_{i+1}$. The saving function is $y_i^i - c_i^i = s(\alpha_i; y_i^i, y_{i+1}^i)$ with $\alpha_i = R_i^{-1}$ (9.4.5). Equilibrium requires $M/p_i = s(\cdot)$ (9.4.6): the saving of the young equals the real balances the old sell. With log utility, $s = .5(w_1 - w_2/R)$. This gives $p_t = 2M/w_1 + (w_2/w_1)p_{t+1}$ (9.4.7) and the continuum $p_t = 2M/[w_1(1 - w_2/w_1)] + c(w_1/w_2)^t$, $c \geq 0$ (9.4.8).

Deficit finance (section 9.5). With $f(R_t) = M_t/p_t$ and $d = g - \tau_1 - \tau_2$, we get $f(R_t) = f(R_{t-1})R_{t-1} + d$ (9.5.4a). Stationary equilibria solve $f(R)(1 - R) = d$ (9.5.5a), a Laffer curve with a high-return and a low-return root.

Optimality (section 9.7). Let $\theta_{\text{aut}} = u_1/u_2$ at the endowment. A monetary equilibrium exists iff $\theta_{\text{aut}}z < n$. Autarky is optimal iff $\theta_{\text{aut}} \geq n$. Given $\theta_{\text{aut}}z < n$, the stationary monetary equilibrium is optimal iff $z \leq 1$. The Balasko–Shell criterion (9.7.18): an allocation is optimal iff $\sum_t \prod_{s \leq t} (1 + r_s) = \infty$.

Within-generation heterogeneity (section 9.8). A nonmonetary equilibrium solves $\sum_j N_j f(R, j) = 0$. A monetary one solves $\sum_j N_j f(R_t, j) = H/p_t$ with $R_t = p_t/p_{t+1}$. Real-bills open-market operations are irrelevant (9.8.6).

Two tools used repeatedly

(T1) Log saving. With $u = \ln c$ and endowment (y_1, y_2) , maximising $\ln(y_1 - s) + \ln(y_2 + Rs)$ gives $\frac{1}{y_1 - s} = \frac{R}{y_2 + Rs}$, so $y_2 + Rs = R(y_1 - s)$ and

$$s(R) = \frac{1}{2} \left(y_1 - \frac{y_2}{R} \right), \quad c_y = \frac{1}{2} \left(y_1 + \frac{y_2}{R} \right), \quad c_o = \frac{1}{2} (Ry_1 + y_2).$$

(T2) Indirect utility. Substituting (T1), $v(R) = \ln c_y + \ln c_o = 2 \ln(Ry_1 + y_2) - \ln R - 2 \ln 2$. Then $v'(R) = \frac{2y_1}{Ry_1 + y_2} - \frac{1}{R} = \frac{Ry_1 - y_2}{R(Ry_1 + y_2)} = \frac{2s(R)}{R(Ry_1 + y_2)}$. So a *saver* ($s > 0$) gains from a higher return and a *borrower* ($s < 0$) loses.

Samuelson Economies with Fiat Currency

p. 366. At each date $t \geq 1$, an economy consists of overlapping generations of a constant number N of two-period-lived agents. Young agents born in t have preferences over consumption streams of a single good that are ordered by $u(c_t^t) + u(c_{t+1}^t)$, where $u(c) = c^{1-\gamma}/(1-\gamma)$, and where c_t^i is the consumption of an agent born at i in time t . It is understood that $\gamma > 0$, and that when $\gamma = 1$, $u(c) = \ln c$. Each young agent born at $t \geq 1$ has identical preferences and endowment pattern (w_1, w_2) , where w_1 is the endowment when young and w_2 is the endowment when old. Assume $0 < w_2 < w_1$. In addition, there are some initial old agents at time 1 who are endowed with w_2 of the time 1 consumption good, and who order consumption streams by c_1^0 . The initial old (i.e., the old at $t = 1$) are also endowed with M units of unbacked fiat currency. The stock of currency is constant over time.

- Find the saving function of a young agent.
- Define an equilibrium with valued fiat currency.
- Define a stationary equilibrium with valued fiat currency.
- Compute a stationary equilibrium with valued fiat currency.
- Describe how many equilibria with valued fiat currency there are. (You are not being asked to compute them.)
- Compute the limiting value as $t \rightarrow +\infty$ of the rate of return on currency in each of the nonstationary equilibria with valued fiat currency. Justify your calculations.

Solution: Let $R_t = p_t/p_{t+1}$ be the gross real return on currency, s_t saving per young agent, and $m_t \equiv M/(Np_t)$ real balances per young agent.

(a) The young agent solves $\max_s u(w_1 - s) + u(w_2 + R_t s)$. Its first-order condition is $u'(w_1 - s) = R_t u'(w_2 + R_t s)$, that is $(w_1 - s)^{-\gamma} = R_t (w_2 + R_t s)^{-\gamma}$. Raise both sides to $-1/\gamma$: $w_1 - s = R_t^{-1/\gamma} (w_2 + R_t s)$. Collect the s terms: $s(1 + R_t^{1-1/\gamma}) = w_1 - R_t^{-1/\gamma} w_2$. So

$$s(R_t) = \frac{w_1 - R_t^{-1/\gamma} w_2}{1 + R_t^{1-1/\gamma}} \quad (\gamma = 1 : s = \frac{1}{2}(w_1 - w_2/R_t)).$$

Money holdings require $s \geq 0$. That holds iff $R_t \geq (w_2/w_1)^\gamma$, the autarky MRS $u'(w_1)/u'(w_2)$.

(b) An *equilibrium with valued fiat currency* is a positive sequence $\{p_t\}_{t \geq 1}$ with $p_t < \infty$ and an allocation $\{c_1^0; (c_t^t, c_{t+1}^t)_{t \geq 1}\}$ such that three conditions hold. (i) Given $\{p_t\}$, $(c_t^t, c_{t+1}^t) = (w_1 - s(R_t), w_2 + R_t s(R_t))$ is optimal for every $t \geq 1$. (ii) The money market clears, $M/p_t = Ns(R_t)$, equivalently $m_t = s(R_t)$ with $R_t = m_{t+1}/m_t$. (iii) The initial old consume $c_1^0 = w_2 + M/(Np_1)$. Goods-market clearing $c_t^t + c_t^{t-1} = w_1 + w_2$ then follows from the budget constraints.

(c) A *stationary equilibrium* has $(c_t^t, c_{t+1}^t) = (c_y, c_o)$ for all $t \geq 1$ (the book's definition, p. 333). With M constant, $m_t = s(R_t)$ constant forces $p_t = p$ and $R_t = 1$.

(d) Setting $R = 1$ in (a) gives $s(1) = (w_1 - w_2)/2$ for every γ . Then

$$p = \frac{2M}{N(w_1 - w_2)}, \quad c_y = c_o = \frac{w_1 + w_2}{2}, \quad c_1^0 = \frac{w_1 + w_2}{2}.$$

Currency is valued because $w_1 > w_2$.

(e) Write the equilibrium condition in consumption terms. We have $c_t^t = w_1 - m_t$ and $c_{t+1}^t = w_2 + m_{t+1}$. The Euler equation with $R_t = m_{t+1}/m_t$ becomes

$$L(m_t) \equiv m_t u'(w_1 - m_t) = m_{t+1} u'(w_2 + m_{t+1}) \equiv \Phi(m_{t+1}).$$

L is strictly increasing. $\Phi'(m) = (w_2 + m)^{-\gamma-1}[w_2 + (1 - \gamma)m]$, which is positive on $[0, m^*]$, $m^* \equiv (w_1 - w_2)/2$, whenever $\gamma \leq 1$ or $(\gamma - 1)m^* < w_2$. On that branch $m_{t+1} = \Phi^{-1}(L(m_t))$ is well defined. The fixed points solve $u'(w_1 - m) = u'(w_2 + m)$ or $m = 0$, so they are $m = m^*$ and $m = 0$. At 0, $L'(0) = u'(w_1) < u'(w_2) = \Phi'(0)$, so $L < \Phi$ on $(0, m^*)$. Hence $\Phi(m_{t+1}) = L(m_t) < \Phi(m_t)$ gives $m_{t+1} < m_t$. Any $m_1 \in (0, m^*)$ generates a positive, decreasing sequence that converges to 0: an equilibrium. Any $m_1 > m^*$ makes m_t increase until $m_t \geq w_1$, which is infeasible. So there is a **continuum** of equilibria with valued currency, indexed by $p_1 \in [p^*, \infty)$. Exactly one is stationary; all the others have $p_t \uparrow \infty$. If γ is so large that Φ bends back on $[0, m^*]$ (a backward-bending offer curve), m_{t+1} can have two solutions, and there are even more equilibria, cycles included.

(f) Along a nonstationary equilibrium $m_t \rightarrow 0$. From the Euler equation,

$$R_t = \frac{u'(w_1 - m_t)}{u'(w_2 + m_{t+1})} \longrightarrow \frac{u'(w_1)}{u'(w_2)} = \left(\frac{w_2}{w_1}\right)^\gamma < 1,$$

the autarkic interest factor. For $\gamma = 1$ this is w_2/w_1 , consistent with (9.4.8), where $p_{t+1}/p_t \rightarrow w_1/w_2$.

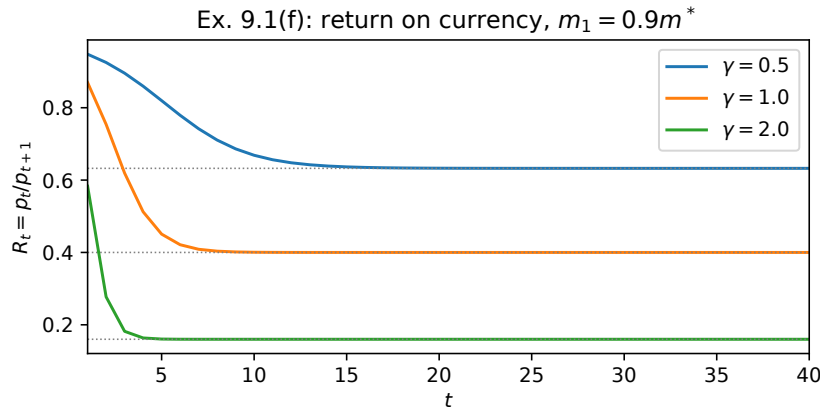


Figure 1: Exercise 9.1(f). R_t along nonstationary equilibria with $m_1 = 0.9m^*$ ($w_1 = 1$, $w_2 = 0.4$). Dotted lines: $(w_2/w_1)^\gamma$.

Numerical exercise — see the companion code

Parameters: $w_1 = 1$, $w_2 = .4$, $\gamma \in \{.5, 1, 2\}$. The closed-form $s(R)$ matches numerical maximisation to 10^{-6} . For every γ , $s(1) = .3 = m^*$ is a fixed point. From $m_1 = .9m^*$ the forward map gives a strictly falling m_t and $R_{80} = .632456, .400000, .160000$, which equals $(w_2/w_1)^\gamma$.

Exercise 9.2 Lenders and borrowers in one generation

Question

p. 367. Consider an economy with overlapping generations of a constant population of an even number N of two-period-lived agents. New young agents are born at each date $t \geq 1$. Half of the young agents are endowed with w_1 when young and 0 when old. The other half are endowed with 0 when young and w_2 when old. Assume $0 < w_2 < w_1$. Preferences of all young agents are as in problem 1, with $\gamma = 1$. Half of the N initial old are endowed with w_2 units of the consumption good and half are endowed with nothing. Each old person orders consumption streams by c_1^0 . Each old person at $t = 1$ is endowed with M units of unbacked fiat currency. No other generation is endowed with fiat currency. The stock of fiat currency is fixed over time.

- a. Find the saving function of each of the two types of young person for $t \geq 1$.
- b. Define an equilibrium without valued fiat currency. Compute all such equilibria.
- c. Define an equilibrium with valued fiat currency.
- d. Compute all the (nonstochastic) equilibria with valued fiat currency.
- e. Argue that there is a unique stationary equilibrium with valued fiat currency.
- f. How are the various equilibria with valued fiat currency ranked by the Pareto criterion?

Solution: Type 1 has endowment $(w_1, 0)$ and type 2 has $(0, w_2)$. R_t is the gross return on consumption loans (and on currency, when valued). Total currency is NM .

(a) By (T1), $s_1(R_t) = w_1/2$ (inelastic) and $s_2(R_t) = -w_2/(2R_t)$ (a borrower).

(b) An *equilibrium without valued currency* is a sequence $\{R_t\}$ and allocations such that each young person optimises given R_t and the loan market clears, $\frac{N}{2}s_1(R_t) + \frac{N}{2}s_2(R_t) = 0$, with $1/p_t = 0$. The clearing condition reads $\frac{N}{2} \cdot \frac{w_1}{2} = \frac{N}{2} \cdot \frac{w_2}{2R_t}$, so $R_t = w_2/w_1$ for every t . The equilibrium is unique because each date is a separate Fisher loan economy. Both types consume $(w_1/2, w_2/2)$: type 1 gets $c_o = R_t w_1/2 = w_2/2$ and type 2 gets $c_y = w_2/(2R_t) = w_1/2$. The initial old consume their endowments, w_2 or 0.

(c) An *equilibrium with valued fiat currency* is a positive sequence $\{p_t\}$, $p_t < \infty$, with $R_t = p_t/p_{t+1}$, such that currency and loans bear the same return, each young person optimises, and the young's net saving equals the old's real balances:

$$\frac{N}{2} [s_1(R_t) + s_2(R_t)] = \frac{NM}{p_t} \iff \frac{1}{4} \left(w_1 - w_2 \frac{p_{t+1}}{p_t} \right) = \frac{M}{p_t}.$$

The initial old consume their goods endowment plus M/p_1 .

(d) Multiply by $4p_t/w_1$: $p_t = \frac{4M}{w_1} + \frac{w_2}{w_1} p_{t+1}$, the analogue of (9.4.7). Let $p^* = 4M/(w_1 - w_2)$ be its fixed point. Then $p_{t+1} - p^* = \frac{w_1}{w_2} (p_t - p^*)$, and

$$p_t = p^* + (p_1 - p^*) \left(\frac{w_1}{w_2} \right)^{t-1}, \quad p_1 \geq p^*.$$

If $p_1 < p^*$, then $p_t \rightarrow -\infty$, which is impossible. Every $p_1 \geq p^*$ is an equilibrium. Along it, $R_t = p_t/p_{t+1} \in [w_2/w_1, 1]$ and $R_t \rightarrow w_2/w_1$ when $p_1 > p^*$. Allocations: type 1 gets $(w_1/2, R_t w_1/2)$ and type 2 gets $(w_2/(2R_t), w_2/2)$.

(e) Stationarity requires constant real balances M/p_t , so $p_t = p$ and $R = 1$. From (d), $p = p^* = 4M/(w_1 - w_2)$, which is unique. Put differently, $p_1 = p^*$ is the only member of the family with $p_{t+1} = p_t$.

(f) Let $x_t = (p_1 - p^*)(w_1/w_2)^{t-1} \geq 0$. Then $R_t = \frac{p^* + x_t}{p^* + x_t w_1/w_2}$, which is strictly decreasing in x_t . So a higher p_1 lowers R_t at *every* date. By (T2), lenders (type 1, $s > 0$) lose and borrowers (type 2, $s < 0$) gain. The initial old, who hold M/p_1 , lose. **The equilibria with valued currency are not Pareto ranked.** Moving to a lower p_1 helps type 1 and the initial old but hurts every type-2 agent. This contrasts with 9.5, where every young agent is a saver.

Numerical exercise — see the companion code

$w_1 = 1$, $w_2 = .5$, $N = 2$, $M = 1$. `brentq` on the loan market gives $R = .5 = w_2/w_1$. The closed-form p_t solves the recursion and clears the money market for $p_1 \in \{p^*, 1.2p^*, 2p^*\}$, $p^* = 8$. Comparing $p_1 = p^*$ with $1.5p^*$: type-1 utility is higher at every t and type-2 utility is lower at every t .

Exercise 9.3 A Lucas tree

Question

p. 368. Take the economy of exercise 9.1, but make one change. Endow the initial old with a tree that yields a constant dividend of $d > 0$ units of the consumption good for each $t \geq 1$.

- Compute all the equilibria with valued fiat currency.
- Compute all the equilibria without valued fiat currency.
- If you want, you can answer both parts of this question in the context of the following particular numerical example: $w_1 = 10$, $w_2 = 5$, $d = .000001$.

Solution: *Reading.* As in Example 3 (p. 338), where the feasibility line is $c_t^i + c_t^{i-1} = y_t^i + y_t^{i-1} + d$, d is the dividend *per young agent* (per capita). Let a_t be the ex-dividend price of the per-capita tree claim at t . Buying it at t pays $a_{t+1} + d$ at $t + 1$. Two conditions hold. Tree arbitrage: $a_t R_t = a_{t+1} + d$. Market clearing: $s(R_t) = a_t + m_t$.

(a) **No equilibrium with valued fiat currency exists.** Suppose currency is valued at all t . (If it were worthless at some $t + 1$, nobody would buy it at t .) Then $R_t = m_{t+1}/m_t$, the tree must earn the same return, and $0 < m_t \leq s(R_t) < w_1$. Iterate the arbitrage condition T times:

$$a_1 = \sum_{j=1}^T \frac{d}{R_1 \cdots R_j} + \frac{a_{T+1}}{R_1 \cdots R_T} \geq \sum_{j=1}^T d \frac{m_1}{m_{j+1}} \geq T d \frac{m_1}{w_1}.$$

Here $R_1 \cdots R_j = m_{j+1}/m_1$ and $a_{T+1} \geq 0$. The right side tends to ∞ as $T \rightarrow \infty$,

but $a_1 \leq s(R_1) < w_1$. This is a contradiction. Valued money would have to keep a nonvanishing share of saving, which forces $\prod R_s$ to stay bounded. Then the tree, with its perpetual dividend, would be worth an unbounded amount.

(b) Without valued currency, $s(R_t) = a_t$ and $a_{t+1} = R_t a_t - d$. A stationary point satisfies $s(R)(R - 1) = d$. Since $d > 0$ and $a = s(R) > 0$ (a positive-dividend asset has a positive price), we need $R > 1$. For log utility, $a = s(R) = \frac{1}{2}(w_1 - w_2/R)$ gives $R = w_2/(w_1 - 2a)$, and the map is

$$a_{t+1} = \varphi(a_t) \equiv \frac{w_2 a_t}{w_1 - 2a_t} - d, \quad \varphi'(a) = \frac{w_1 w_2}{(w_1 - 2a)^2}.$$

Fixed points: $(a + d)(w_1 - 2a) = w_2 a$, so $2a^2 - (w_1 - w_2 - 2d)a - dw_1 = 0$. The product of the roots is $-dw_1/2 < 0$, so there is one positive root $\bar{a}$ and one negative root:

$$\bar{a} = \frac{(w_1 - w_2 - 2d) + \sqrt{(w_1 - w_2 - 2d)^2 + 8dw_1}}{4}, \quad \bar{R} = \frac{\bar{a} + d}{\bar{a}} > 1, \quad \bar{a} = \frac{d}{\bar{R} - 1}.$$

At $\bar{a}$, $\varphi'(\bar{a}) > 1$ ($\bar{a}$ is near $(w_1 - w_2)/2$, so $\varphi' \approx w_1/w_2$). If $a_1 > \bar{a}$, the path rises until $a_t \geq w_1/2$, which is infeasible. If $a_1 < \bar{a}$, it falls through 0 toward the negative root, which is also infeasible. **The unique equilibrium is stationary:** $a_t = \bar{a}$, $R_t = \bar{R} > 1$. It is the book's Example 3 (Figure 9.2.3). For general γ the same graph applies: the candidate with $R < 1$ gives the old unbounded wealth.

(c) With $w_1 = 10$, $w_2 = 5$, $d = 10^{-6}$ and $\gamma = 1$: $\bar{a} = 2.5000010$, $\bar{R} - 1 = 4.0 \times 10^{-7}$, $c_y = w_1 - \bar{a} = 7.4999990$, $c_o = w_2 + \bar{a} + d \approx 7.5$. A microscopic dividend is enough to eliminate every monetary equilibrium and all the equilibria that converge to autarky. It selects (almost exactly) the golden-rule allocation $c_y = c_o = 7.5$, with the tree playing the role that fiat money plays in the stationary equilibrium of 9.1(d).

Numerical exercise — see the companion code

`brentq` on $s(R)(R - 1) = d$ equals the quadratic root ($\gamma = 1$), and gives $R - 1 = 4.0 \times 10^{-7}$, $a = 2.5000002$ for $\gamma = 2$. Paths from $\bar{a}(1 \pm 10^{-4})$ leave $(0, w_1/2)$ at $t = 14$ (above) and $t = 35$ (below). For (a), the bound Tdm_1/w_1 exceeds w_1 for large finite T .

Exercise 9.4 Population growth

Question

p. 368. Take the economy of exercise 9.1 and make the following two changes. First, assume that $\gamma = 1$. Second, assume that the number of young agents born at t is $N(t) = nN(t-1)$, where $N(0) > 0$ is given and $n \geq 1$. Everything else about the economy remains the same.

- a. Compute an equilibrium without valued fiat money.
- b. Compute a stationary equilibrium with valued fiat money.

Solution:

(a) With $1/p_t = 0$ and identical young agents, there is no trade: $c_t^t = w_1$, $c_{t+1}^t = w_2$, $c_1^0 = w_2$. The shadow interest factor that makes the young content with zero saving, $s(R) = 0$, is $R = w_2/w_1 < 1$.

(b) Money-market clearing is $N(t)s(R_t) = M/p_t$. Let $m_t = M/(N(t)p_t)$ be real balances per young person. Stationarity means $m_t = m$. Then $p_t N(t)$ is constant, so $p_{t+1}/p_t = N(t)/N(t+1) = 1/n$ and $\boxed{R_t = n}$ (the high-interest-rate equilibrium). By (T1),

$$m = s(n) = \frac{1}{2} \left(w_1 - \frac{w_2}{n} \right) > 0, \quad p_t = \frac{M}{N(0)n^t s(n)} = \frac{2Mn^{1-t}}{N(0)(nw_1 - w_2)},$$

$$c_y = \frac{1}{2} \left(w_1 + \frac{w_2}{n} \right), \quad c_o = w_2 + nm = \frac{1}{2}(nw_1 + w_2), \quad c_1^0 = w_2 + \frac{M}{N(0)p_1} = w_2 + nm.$$

$s(n) > 0$ because $nw_1 \geq w_1 > w_2$. Feasibility per young person, $c_y + c_o/n = w_1 + w_2/n$, holds. The price level falls at rate n , so currency pays the population growth rate.

Numerical exercise — see the companion code

$w_1 = 1$, $w_2 = .4$, $n = 1.1$: $s(n) = 0.318182$, $(c_y, c_o) = (0.681818, 0.75)$. We verify $R_t = n$ and market clearing at every date, the feasibility identity, and that the closed-form $s(n)$ equals the numerical optimum.

Exercise 9.5 All monetary equilibria with growth

Question

p. 368. Consider an economy consisting of overlapping generations of two-period-lived consumers. At each date $t \geq 1$ there are born $N(t)$ identical young people each of whom is endowed with $w_1 > 0$ units of a single consumption good when young and $w_2 > 0$ units of the consumption good when old. Assume that $w_2 < w_1$. The consumption good is not storable. The population of young people is described by $N(t) = nN(t-1)$, where $n > 0$. Young people born at t rank utility streams according to $\ln(c_t^i) + \ln(c_{t+1}^i)$ where c_t^i is the consumption of the time t good of an agent born in i . In addition, there are $N(0)$ old people at time 1, each of whom is endowed with w_2 units of the time 1 consumption good. The old at $t = 1$ are also endowed with one unit of unbacked pieces of infinitely durable but intrinsically worthless pieces of paper called fiat money.

- a. Define an equilibrium without valued fiat currency. Compute such an equilibrium.
- b. Define an equilibrium with valued fiat currency.
- c. Compute all equilibria with valued fiat currency.
- d. Find the limiting rates of return on currency as $t \rightarrow +\infty$ in each of the equilibria that you found in part c. Compare them with the one-period interest rate in the equilibrium in part a.
- e. Are the equilibria in part c ranked according to the Pareto criterion?

Solution: *Reading.* “One unit” is per initial old person, so the currency stock is $N(0)$.

(a) An *equilibrium without valued currency* is $1/p_t = 0$ with each young agent optimising given the loan rate R_t , and zero aggregate saving, $N(t)s(R_t) = 0$. Since the agents are identical, $s(R_t) = 0$, so $R_t = w_2/w_1$ (autarky: $c_t^i = w_1$, $c_{t+1}^i = w_2$, $c_1^0 = w_2$).

(b) An *equilibrium with valued currency* is a positive sequence $\{p_t\}$, $p_t < \infty$, such that, with $R_t = p_t/p_{t+1}$, each young agent saves $s(R_t) = \frac{1}{2}(w_1 - w_2/R_t)$ and $N(t)s(R_t) = N(0)/p_t$. Consumption is $c_t^i = w_1 - s(R_t)$, $c_{t+1}^i = w_2 + R_t s(R_t)$, and $c_1^0 = w_2 + 1/p_1$.

(c) Multiply the clearing condition by $2p_t/N(t)$, using $N(0)/N(t) = n^{-t}$: $2n^{-t} = w_1 p_t - w_2 p_{t+1}$, that is

$$p_{t+1} = \frac{w_1}{w_2} p_t - \frac{2}{w_2} n^{-t}.$$

Particular solution. Try $p_t = K n^{-t}$: $K n^{-t-1} = \frac{w_1}{w_2} K n^{-t} - \frac{2}{w_2} n^{-t}$, so $K \left(\frac{w_1}{w_2} - \frac{1}{n} \right) = \frac{2}{w_2}$ and $K = \frac{2n}{nw_1 - w_2}$. *General solution:*

$$p_t = \frac{2n}{nw_1 - w_2} n^{-t} + c \left(\frac{w_1}{w_2} \right)^t.$$

Case $nw_1 > w_2$ ($n > w_2/w_1 = \theta_{\text{aut}}$). Here $K > 0$ and $w_1/w_2 > 1/n$. If $c < 0$, the second term dominates and $p_t < 0$ eventually. So the equilibria are exactly $c \geq 0$ (a continuum), with $c = 0$ the stationary one: real balances per young $s(n)$ constant. *Case $nw_1 \leq w_2$.* If $nw_1 < w_2$, then $K < 0$ and $1/n > w_1/w_2$, so the first term dominates and $p_t < 0$

eventually for every c . If $nw_1 = w_2$, write $p_t = q_t n^{-t}$; then $q_{t+1} = q_t - 2n/w_2 \rightarrow -\infty$. Either way there is **no** monetary equilibrium. This is the proposition of section 9.7 with $z = 1$: one exists iff $\theta_{\text{aut}} < n$.

(d) For $c = 0$, $R_t = n$ for all t . For $c > 0$, $p_t \sim c(w_1/w_2)^t$, so $R_t = p_t/p_{t+1} \rightarrow w_2/w_1$. That is the one-period interest factor of the nonmonetary equilibrium in (a). The nonstationary equilibria are asymptotically autarkic.

(e) **Yes, they are completely ranked.** Write $a_t = Kn^{-t} > 0$ and $b_t = c(w_1/w_2)^t \geq 0$. Then

$$R_t = \frac{a_t + b_t}{a_t/n + b_t w_1/w_2},$$

a weighted average (in the sense of a ratio of sums) of n and w_2/w_1 . Its weight shifts toward $w_2/w_1 < n$ as b_t/a_t rises, and b_t/a_t is increasing in c . So a higher c lowers R_t at every t . Every young agent is a saver ($s(R_t) = m_t > 0$), so by (T2) $v(R_t)$ falls: every generation $t \geq 1$ is worse off. The initial old consume $w_2 + 1/p_1$, which falls with c . The stationary equilibrium ($c = 0$) Pareto dominates every other, and a lower c dominates a higher c . Since $n > \theta_{\text{aut}}$ and $z = 1$, the stationary one is Pareto optimal (section 9.7). The others converge to the nonoptimal autarky.

Numerical exercise — see the companion code

$w_1 = 1$, $w_2 = .5$, $n = .9$ ($K = 4.5$). For $c \in \{0, 10^{-6}, 10^{-3}, 1\}$ the closed form solves the recursion and clears the money market. R_{200} equals n ($c = 0$) or w_2/w_1 ($c > 0$). Utilities of all 199 generations and $1/p_1$ decrease with c . With $n = .4 < w_2/w_1$, every $c \geq 0$ gives $p_t < 0$ within 60 periods.

Several Currencies, Credit, and Inside Money

Exercise 9.6 Exchange rate determinacy

Question

pp. 369–370. The world consists of two economies, named $i = 1, 2$, which except for their governments' policies are “copies” of one another. At each date $t \geq 1$, there is a single consumption good, which is storable, but only for rich people. Each economy consists of overlapping generations of two-period-lived agents. For each $t \geq 1$, in economy i , N poor people and N rich people are born. Let $c_t^h(s), y_t^h(s)$ be the time s (consumption, endowment) of a type h agent born at t . Poor agents are endowed with $[y_t^h(t), y_t^h(t+1)] = (\alpha, 0)$; rich agents are endowed with $[y_t^h(t), y_t^h(t+1)] = (\beta, 0)$, where $\beta \gg \alpha$. In each country, there are $2N$ initial old who are endowed in the aggregate with $H_i(0)$ units of an unbacked currency and with $2N\epsilon$ units of the time 1 consumption good. For the rich people, storing k units of the time t consumption good produces Rk units of the time $t + 1$ consumption good, where $R > 1$ is a fixed gross rate of return on storage. Rich people can earn the rate of return R either by storing goods or by lending to either government by means of indexed bonds. We assume that poor people are prevented from storing capital or holding indexed government debt by the sort of denomination and intermediation restrictions described by Sargent and Wallace (1982).

For each $t \geq 1$, all young agents order consumption streams according to $\ln c_t^h(t) + \ln c_t^h(t+1)$.

For $t \geq 1$, the government of country i finances a stream of purchases (to be thrown into the ocean) of $G_i(t)$ subject to the following budget constraint:

$$(1) \quad G_i(t) + RB_i(t-1) = B_i(t) + \frac{H_i(t) - H_i(t-1)}{p_i(t)} + T_i(t),$$

where $B_i(0) = 0$; $p_i(t)$ is the price level in country i ; $T_i(t)$ are lump-sum taxes levied by the government on the *rich* young people at time t ; $H_i(t)$ is the stock of i 's fiat currency at the end of period t ; $B_i(t)$ is the stock of indexed government interest-bearing debt (held by the rich of either country). The government does not explicitly tax poor people, but might tax through an inflation tax. Each government levies a lump-sum tax of $T_i(t)/N$ on each young rich citizen of its own country.

Poor people in both countries are free to hold whichever currency they prefer. Rich people can hold debt of either government and can also store; storage and both government debts bear a constant gross rate of return R .

- a. Define an *equilibrium* with valued fiat currencies (in both countries).
- b. In a nonstochastic equilibrium, verify the following proposition: if an equilibrium exists in which both fiat currencies are valued, the exchange rate between the two currencies must be constant over time.
- c. Suppose that government policy in each country is characterized by specified (exogenous) levels $G_i(t) = G_i$, $T_i(t) = T_i$, $B_i(t) = 0$, $\forall t \geq 1$. (The remaining elements of government policy adjust to satisfy the government budget constraints.) Assume that the exogenous components of policy have been set so that an equilibrium with two valued fiat currencies exists. Under this description of policy, show that the equilibrium exchange rate is indeterminate.

d. Suppose that government policy in each country is described as follows: $G_i(t) = G_i$, $T_i(t) = T_i$, $H_i(t+1) = H_i(1)$, $B_i(t) = B_i(1) \forall t \geq 1$. Show that if there exists an equilibrium with two valued fiat currencies, the exchange rate is determinate.

e. Suppose that government policy in country 1 is specified in terms of exogenous levels of $s_1 = [H_1(t) - H_1(t-1)]/p_1(t) \forall t \geq 2$, and $G_1(t) = G_1 \forall t \geq 1$. For country 2,

Solution: Write $R_{it} = p_i(t)/p_i(t+1)$ for the gross return on currency i (distinct from the storage return R), $h_i(t) = H_i(t)/p_i(t)$ for its real stock, and $e(t) = p_1(t)/p_2(t)$ for the exchange rate. By (T1) with endowment $(y, 0)$, each young person saves half the after-tax endowment, whatever the return. The $2N$ poor of the world therefore demand $N\alpha$ of real balances in total. The rich save $(\beta - T_i/N)/2$ each, in storage or bonds at R .

(a) An *equilibrium with two valued currencies* consists of positive finite sequences $\{p_1(t), p_2(t)\}$, stocks $\{H_i(t), B_i(t)\}$, storage, and allocations, for given policy, such that five conditions hold. (i) The poor maximise: $c^{\text{poor}}(t) = \alpha/2$, and they hold only currencies with the highest return, $\max_i R_{it}$. (ii) The rich maximise: they save $(\beta - T_i(t)/N)/2$ in storage and bonds, and hold currency only if its return is at least R . (iii) The world currency market clears, $h_1(t) + h_2(t) = N\alpha$, with $h_i(t) > 0$ for both i . (iv) The rich's saving equals storage plus $B_1(t) + B_2(t)$. (v) Both budget constraints (1) hold, and goods markets clear.

(b) Both currencies are held by the poor at every t . If $R_{1t} > R_{2t}$, no one would hold currency 2 at t , so $h_2(t) = 0$, a contradiction. Hence $R_{1t} = R_{2t}$, that is $\frac{p_1(t)}{p_1(t+1)} = \frac{p_2(t)}{p_2(t+1)}$.

Rearranged, $\frac{p_1(t+1)}{p_2(t+1)} = \frac{p_1(t)}{p_2(t)}$, so $e(t+1) = e(t) = e$.

(c) With $B_i = 0$, (1) gives $[H_i(t) - H_i(t-1)]/p_i(t) = G_i - T_i \equiv d_i$ for every $t \geq 1$. Let R_m be the common currency return. Then $h_i(t) = \frac{H_i(t-1)}{p_i(t-1)} \frac{p_i(t-1)}{p_i(t)} + d_i = R_m h_i(t-1) + d_i$ for $t \geq 2$. Summing over i and using (iii): $N\alpha = R_m N\alpha + d_1 + d_2$, so $R_m = 1 - (d_1 + d_2)/(N\alpha)$, a constant. At $t = 1$, with $p_2(1) = p_1(1)/e$, the two budgets add to

$$\frac{H_1(0) + e H_2(0)}{p_1(1)} + d_1 + d_2 = N\alpha.$$

This is *one* equation in two unknowns $(p_1(1), e)$. Pick any $e > 0$. Then $p_1(1) = \frac{H_1(0) + e H_2(0)}{N\alpha - d_1 - d_2}$, $p_i(t) = p_i(1) R_m^{t-1}$, and $h_i(t) = R_m^{t-1} [h_i(1) - \bar{h}_i] + \bar{h}_i > 0$ with $\bar{h}_i = d_i/(1 - R_m)$. Every condition of (a) holds: the poor consume $(\alpha/2, R_m \alpha/2)$, the rich and the governments are unaffected, and $R_m < R$. So **every** $e \in (0, \infty)$ **is an equilibrium exchange rate** (Kareken–Wallace). The real allocation is the same for all e . Only the split of world real balances between the two currencies and between the two countries' initial old depends on e . *Remark.* Only one e keeps each currency's real stock constant, $e = \frac{H_1(0)d_2}{H_2(0)d_1}$ (needs $d_1, d_2 > 0$). This is the book's section 27.3.11 selection; the definition in (a) does not impose it.

(d) Now the nominal stocks $H_i(t) = H_i(1)$ and bonds $B_i(1)$ are policy. For $t \geq 2$, (1) reads $G_i + R B_i(1) = B_i(1) + 0 + T_i$, a restriction on policy: $(R - 1)B_i(1) = T_i - G_i$. With e constant and both stocks constant, (iii) gives $\frac{H_1(1) + e H_2(1)}{p_1(t)} = N\alpha$, so $p_1(t)$ and $p_2(t)$ are constant for $t \geq 1$ and $R_m = 1$. The $t = 1$ budget, $G_i = B_i(1) + [H_i(1) - H_i(0)]/p_i(1) + T_i$,

now pins each price level separately:

$$S_i \equiv G_i - T_i - B_i(1) = \frac{R}{R-1}(G_i - T_i), \quad p_i(1) = \frac{H_i(1) - H_i(0)}{S_i},$$

$$e = \frac{[H_1(1) - H_1(0)] S_2}{[H_2(1) - H_2(0)] S_1}.$$

Existence requires the policies to satisfy $H_1(1)/p_1 + H_2(1)/p_2 = N\alpha$; that is the “if there exists” clause. If one country has $S_i = 0$ (so $H_i(1) = H_i(0)$), its price level is still determined, by (iii) given the other’s. Only when both have $S_i = 0$ do we return to indeterminacy. The difference from (c) is that the governments now fix *nominal* quantities, which anchor each currency’s value through its own budget constraint.

(e) *Reading.* The statement leaves some instruments unnamed. We keep $T_i(t) = T_i$ from (c)–(d) in both countries, and let each country’s remaining element adjust. For country 1 that is B_1 , subject to boundedness: the rich’s saving is bounded. For country 2 it is currency, as a one-time nominal issue $H_2(1)$ that is then held fixed (as in (d)). *Country 2.* For $t \geq 2$, (1) requires $T_2 = G_2 + (R-1)B_2(1)$. At $t = 1$, $p_2(1) = [H_2(1) - H_2(0)]/(G_2 - T_2 - B_2(1))$ is pinned, exactly as in (d). *Country 1.* For $t \geq 2$, $B_1(t) = RB_1(t-1) + G_1 - T_1 - s_1$. Its solution is $B_1(t) = \bar{b} + R^{t-1}(B_1(1) - \bar{b})$ with $\bar{b} = (T_1 + s_1 - G_1)/(R-1)$. Since $R > 1$, boundedness forces $B_1(1) = \bar{b}$, and then the $t = 1$ seigniorage is pinned too: $\sigma_1 \equiv [H_1(1) - H_1(0)]/p_1(1) = G_1 - T_1 - \bar{b}$. *World money market.* For $t \geq 2$, $N\alpha = R_m N\alpha + s_1$, so $R_m = 1 - s_1/(N\alpha)$. At $t = 1$,

$$\frac{H_1(0)}{p_1(1)} + \sigma_1 + \frac{H_2(1)}{p_2(1)} = N\alpha \implies p_1(1) = \frac{H_1(0)}{N\alpha - \sigma_1 - H_2(1)/p_2(1)},$$

which is unique. Hence $e = p_1(1)/p_2(1)$ is **determinate**, and constant by (b). Country 2’s nominal anchor fixes p_2 , and world money demand then fixes p_1 . *Caveat.* If country 2 instead financed $G_2 + (R-1)B_2$ by money creation every period with T_2 fixed, both governments would be setting real seigniorage. The $t = 1$ condition would again be one equation in $(p_1(1), e)$, as in (c), and e would be indeterminate. Determinacy needs at least one nominal anchor.

Numerical exercise — see the companion code

(c) with $N\alpha = 1$, $H_1(0) = 2$, $H_2(0) = 3$, $d_1 = .05$, $d_2 = .10$: for $e \in \{0.2, 1, 5\}$ the world market clears at every date, $R_m = .85$ for all e , and both real stocks stay positive and converge to $d_i/(1 - R_m)$. $e = 4/3$ gives the constant split. (d) with $H_i(1) = 1.1H_i(0)$, $S_1 = .04$, $S_2 = .06$: $p_1 = p_2 = 5$, $e = 1$, and existence requires $N\alpha = 1.1$.

Exercise 9.7 Credit controls

Question

pp. 370–371. Consider the following overlapping generations model. At each date $t \geq 1$ there appear N two-period-lived young people, said to be of generation t , who live and consume during periods t and $(t + 1)$. At time $t = 1$ there exist N old people who are endowed with $H(0)$ units of paper “dollars,” which they offer to supply inelastically to the young of generation 1 in exchange for goods. Let $p(t)$ be the price of the one good in the model, measured in dollars per time t good. For each $t \geq 1$, $N/2$ members of generation t are endowed with $y > 0$ units of the good at t and 0 units at $(t + 1)$, whereas the remaining $N/2$ members of generation t are endowed with 0 units of the good at t and $y > 0$ units when they are old. All members of all generations have the same utility function:

$$u[c_t^h(t), c_t^h(t + 1)] = \ln c_t^h(t) + \ln c_t^h(t + 1),$$

where $c_t^h(s)$ is the consumption of agent h of generation t in period s . The old at $t = 1$ simply maximize $c_0^h(1)$. The consumption good is nonstorable. The currency supply is constant through time, so $H(t) = H(0)$, $t \geq 1$.

- a. Define a competitive equilibrium without valued currency for this model. Who trades what with whom?
- b. In the equilibrium without valued fiat currency, compute competitive equilibrium values of the gross return on consumption loans, the consumption allocation of the old at $t = 1$, and that of the “borrowers” and “lenders” for $t \geq 1$.
- c. Define a competitive equilibrium with valued currency. Who trades what with whom?
- d. Prove that for this economy there does not exist a competitive equilibrium with valued currency.
- e. Now suppose that the government imposes the restriction that $l_t^h(t)[1 + r(t)] \geq -y/4$, where $l_t^h(t)[1 + r(t)]$ represents claims on $(t + 1)$ -period consumption purchased (if positive) or sold (if negative) by household h of generation t . This is a restriction on the amount of borrowing. For an equilibrium without valued currency, compute the consumption allocation and the gross rate of return on consumption loans.
- f. In the setup of part e, show that there exists an equilibrium with valued currency in which the price level obeys the quantity theory equation $p(t) = qH(0)/N$. Find a formula for the undetermined coefficient q . Compute the consumption allocation and the equilibrium rate of return on consumption loans.
- g. Are lenders better off in economy b or economy f? What about borrowers? What about the old of period 1 (generation 0)?

Solution: “Lenders” have $(y, 0)$ and “borrowers” have $(0, y)$. Let $l_t^h(t)$ be loans made (negative if borrowed). By (T1), a lender saves $y/2$ and an unconstrained borrower sets $l = -y/(2[1 + r(t)])$, promising $y/2$ whatever the rate.

(a) A competitive equilibrium without valued currency is $1/p(t) = 0$ and a sequence

$\{r(t)\}$ with allocations such that each young agent maximises utility given $1 + r(t)$, and the loan market clears, $\frac{N}{2}l^{\text{lend}}(t) + \frac{N}{2}l^{\text{borr}}(t) = 0$. Young lenders trade time- t goods with young borrowers of the *same* generation, for claims on time- $(t + 1)$ goods. There is no trade across generations. The initial old have no goods and worthless paper.

(b) Clearing: $\frac{N}{2} \cdot \frac{y}{2} = \frac{N}{2} \cdot \frac{y}{2[1+r(t)]}$, so $\boxed{1 + r(t) = 1}$. Lenders consume $(y/2, y/2)$. Borrowers consume $c(t) = y/2$ and $c(t + 1) = y - y/2 = y/2$. The initial old consume $c_0^h(1) = 0$.

(c) A *competitive equilibrium with valued currency* is a positive finite sequence $\{p(t)\}$ with $1 + r(t) = p(t)/p(t + 1)$ (currency and loans are perfect substitutes) such that agents optimise and $\frac{N}{2}l^{\text{lend}}(t) + \frac{N}{2}l^{\text{borr}}(t) = H(0)/p(t)$. The old sell dollars to the young (both generations trade), and lenders also lend to borrowers.

(d) Let $m_t = H(0)/p(t)$. Clearing gives $m_t = \frac{Ny}{4} \left[1 - \frac{1}{1+r(t)}\right]$, so $m_t > 0$ requires $1 + r(t) > 1$. With $1 + r(t) = m_{t+1}/m_t$, let $x_t = 4m_t/(Ny) \in (0, 1)$. Then $1 + r(t) = 1/(1 - x_t)$ and $x_{t+1} = x_t/(1 - x_t)$, that is

$$\frac{1}{x_{t+1}} = \frac{1}{x_t} - 1 \quad \implies \quad \frac{1}{x_t} = \frac{1}{x_1} - (t - 1),$$

which is ≤ 1 for $t \geq 1/x_1$. So $x_t \geq 1$ (or negative), which contradicts $x_t < 1$. **No equilibrium with valued currency exists.** This matches section 9.7. The nonmonetary equilibrium has $1 + r = 1 = n$, so it is already optimal ($\theta_{\text{aut}} = n$). Money would need a return above 1 forever, which requires ever-growing real balances.

(e) The borrower wants to promise $y/2$ but may promise only $y/4$. The constraint binds at any rate: borrowing is $y/(4[1 + r])$. Clearing: $\frac{N}{2} \cdot \frac{y}{2} = \frac{N}{2} \cdot \frac{y}{4[1+r]}$, so $\boxed{1 + r(t) = \frac{1}{2}}$. Lenders consume $(y/2, y/4)$. Borrowers consume $(\frac{y}{4}, y - \frac{y}{4}) = (y/4, 3y/4)$. The initial old consume 0.

(f) Try $p(t) = p$ constant, so $1 + r(t) = 1$. Lenders save $y/2$. Borrowers are still constrained at $1 + r = 1$ (they want $y/2 > y/4$), so they borrow $y/4$. Net saving of the young equals real balances:

$$\frac{N}{2} \left(\frac{y}{2} - \frac{y}{4} \right) = \frac{Ny}{8} = \frac{H(0)}{p} \quad \implies \quad p = \frac{8H(0)}{Ny} = q \frac{H(0)}{N}, \quad \boxed{q = \frac{8}{y}}.$$

The allocation is: lenders $(y/2, y/2)$, borrowers $(y/4, 3y/4)$, each initial old $H(0)/(Np) = y/8$. The return on consumption loans is $1 + r = 1$. Check at $t \geq 2$: $\frac{N}{2}(\frac{y}{2} + \frac{y}{4}) + \frac{N}{2}(\frac{y}{2} + \frac{3y}{4}) = Ny$. At $t = 1$: $\frac{N}{2} \cdot \frac{3y}{4} + N\frac{y}{8} = \frac{N}{2}y$. Both equal the endowment. ✓

(g) Lenders consume $(y/2, y/2)$ in both b and f, so they are *indifferent*. Borrowers get $2 \ln y - 2 \ln 2$ in b and $2 \ln y + \ln \frac{3}{16}$ in f. Since $\frac{3}{16} < \frac{1}{4}$, they are *worse off* in f. The initial old get 0 in b and $y/8$ in f, so they are *better off*. The credit control and valued money together transfer resources from borrowers to generation 0.

Numerical exercise — see the companion code

$y = 1$, $N = 2$, $H(0) = 1$. `brentq` gives $1 + r = 1$ (b) and $\frac{1}{2}$ (e). The iteration $x \mapsto x/(1 - x)$ leaves $(0, 1)$ from $x_1 \in \{10^{-3}, .1, .5\}$. We check $q = 8$, feasibility at $t = 1$ and $t \geq 2$, and the three welfare comparisons.

Exercise 9.8 Inside money and real bills

Question

pp. 371–373. Consider the following overlapping generations model of two-period-lived people. At each date $t \geq 1$ there are born N_1 individuals of type 1 who are endowed with $y > 0$ units of the consumption good when they are young and zero units when they are old; there are also born N_2 individuals of type 2 who are endowed with zero units of the consumption good when they are young and $Y > 0$ units when they are old. The consumption good is nonstorable. At time $t = 1$, there are N old people, all of the same type, each endowed with zero units of the consumption good and H_0/N units of unbacked paper called “fiat currency.” The populations of type 1 and 2 individuals, N_1 and N_2 , remain constant for all $t \geq 1$. The young of each generation are identical in preferences and maximize the utility function $\ln c_t^h(t) + \ln c_t^h(t+1)$ where $c_t^h(s)$ is consumption in the s th period of a member h of generation t .

a. Consider the equilibrium without valued currency (that is, the equilibrium in which there is no trade between generations). Let $[1 + r(t)]$ be the gross rate of return on consumption loans. Find a formula for $[1 + r(t)]$ as a function of N_1, N_2, y , and Y .

b. Suppose that N_1, N_2, y , and Y are such that $[1 + r(t)] > 1$ in the equilibrium without valued currency. Then prove that there can exist no quantity-theory-style equilibrium where fiat currency is valued and where the price level $p(t)$ obeys the quantity theory equation $p(t) = q \cdot H_0$, where q is a positive constant and $p(t)$ is measured in units of currency per unit good.

c. Suppose that N_1, N_2, y , and Y are such that in the nonvalued-currency equilibrium $1 + r(t) < 1$. Prove that there exists an equilibrium in which fiat currency is valued and that there obtains the quantity theory equation $p(t) = q \cdot H_0$, where q is a constant. Construct an argument to show that the equilibrium with valued currency is not Pareto superior to the nonvalued-currency equilibrium.

d. Suppose that N_1, N_2, y , and Y are such that, in the preceding nonvalued-currency economy, $[1 + r(t)] < 1$, there exists an equilibrium in which fiat currency is valued. Let $\bar{p}$ be the stationary equilibrium price level in that economy. Now consider an alternative economy, identical with the preceding one in all respects except for the following feature: a government each period purchases a constant amount L_g of consumption loans and pays for them by issuing debt on itself, called “inside money” M_I , in the amount $M_I(t) = L_g \cdot p(t)$. The government never retires the inside money, using the proceeds of the loans to finance new purchases of consumption loans in subsequent periods. The quantity of outside money, or currency, remains H_0 , whereas the “total high-power money” is now $H_0 + M_I(t)$.

(i) Show that in this economy there exists a valued-currency equilibrium in which the price level is constant over time at $p(t) = \bar{p}$, or equivalently, with $\bar{p} = qH_0$ where q is defined in part c.

(ii) Explain why government purchases of private debt are not inflationary in this economy.

(iii) In many models, once-and-for-all government open-market operations in private debt normally affect real variables and/or price level. What accounts for the difference between those models and the one in this exercise?

Solution: *Reading.* N (initial old) is not otherwise defined; its value plays no role. By (T1), type 1 saves $y/2$ and type 2 borrows $Y/(2[1+r])$. Aggregate saving is $S(R) = \frac{1}{2}(N_1y - N_2\frac{Y}{R})$, $R = 1+r$.

(a) $S(R) = 0$ gives $1+r(t) = \frac{N_2Y}{N_1y}$.

(b) Suppose $p(t) = qH_0$ for all t . Then currency, and hence loans, return $p(t)/p(t+1) = 1$. Clearing requires $S(1) = H_0/p = 1/q > 0$. But $S(1) = \frac{1}{2}(N_1y - N_2Y)$, and $1+r = N_2Y/(N_1y) > 1$ means $N_1y < N_2Y$, so $S(1) < 0$. This is a contradiction, so no such equilibrium exists.

(c) Now $N_1y > N_2Y$, so $S(1) > 0$. Set $p(t) = qH_0$ with $q = \frac{2}{N_1y - N_2Y}$. Then $R = 1$

and $S(1) = H_0/p(t)$ at every date, which is an equilibrium. Consumption at $R = 1$: type 1 gets $(y/2, y/2)$, type 2 gets $(Y/2, Y/2)$, and the initial old share $1/q$. *Not Pareto superior:* by (T2), a type-2 agent's utility $\ln \frac{Y}{2R} + \ln \frac{Y}{2}$ is strictly decreasing in R . The monetary equilibrium raises R from $N_2Y/(N_1y) < 1$ to 1, so every type-2 agent of every generation is strictly worse off. Type 1 and the initial old gain. The two equilibria are Pareto non-comparable.

(d)(i) The government lends L_g each period and finances it with $M_I(t) = L_g p(t)$. Lenders' saving is held as outside money, inside money, and private loans. Borrowers borrow from private lenders and from the government. So the loan market condition becomes (the analogue of (9.8.3))

$$\frac{1}{2}N_1y = \frac{H_0 + M_I(t)}{p(t)} + \left(\frac{1}{2}N_2\frac{Y}{R} - L_g\right) \iff S(R_t) = \frac{H_0}{p(t)} + \underbrace{\frac{M_I(t)}{p(t)}}_{=0} - L_g.$$

At $p(t) = \bar{p} = qH_0$, $R = 1$ and $S(1) = H_0/\bar{p}$ holds by (c). The government's budget: it collects $RL_g = L_g$ at $t+1$ and relends it, and $M_I = L_g\bar{p}$ stays constant, so no retirement is needed. Hence $p(t) = \bar{p}$ is an equilibrium, and the allocation is the same as in (c).

(ii) Each dollar of inside money is backed one-for-one by a real loan of equal value and equal return. The public holds L_g more of real balances and L_g less of private loans. Net saving available to absorb *outside* money, $S(R)$, is unchanged, and so is p . This is the irrelevance of open-market operations (9.8.4)–(9.8.6), Adam Smith's real bills doctrine.

(iii) Here currency and private loans are *perfect substitutes*: same return, no legal restrictions, no reserve requirements, no denomination barriers. The government's asset swap is then undone one-for-one by private portfolios (a Modigliani–Miller result for the government's balance sheet, Wallace 1981). Open-market operations matter in models where (1) legal restrictions separate the money and credit markets (Sargent–Wallace 1982), so that money pays less than loans; or (2) the assets bought are not perfect substitutes for money (liquidity, risk, cash-in-advance constraints); or (3) the operation changes the path of taxes or the backing of money, for instance when the government buys its own interest-bearing debt that must be serviced by future taxes.

Numerical exercise — see the companion code

$N_1 = 3$, $N_2 = 2$, $y = Y = 1$, $H_0 = 1$, $L_g = .1$: $1 + r = 2/3$ by `brentq` and by formula, $q = 2$. Type-2 utility is lower at $R = 1$. With inside money, lenders' saving $= (H_0 + M_I)/\bar{p}$ + private loans exactly at $\bar{p}$.

Transfers, Seigniorage, and Deficit Finance**Exercise 9.9 Social security and the price level***Question*

p. 373. Consider an economy (“economy I”) that consists of overlapping generations of two-period-lived people. At each date $t \geq 1$ there is born a constant number N of young people, who desire to consume both when they are young, at t , and when they are old, at $(t + 1)$. Each young person has the utility function $\ln c_t(t) + \ln c_t(t + 1)$, where $c_s(t)$ is time t consumption of an agent born at s . For all dates $t \geq 1$, young people are endowed with $y > 0$ units of a single nonstorable consumption good when they are young and zero units when they are old. In addition, at time $t = 1$ there are N old people endowed in the aggregate with H units of unbacked fiat currency. Let $p(t)$ be the nominal price level at t , denominated in dollars per time t good.

a. Define and compute an equilibrium with valued fiat currency for this economy. Argue that it exists and is unique. Now consider a second economy (“economy II”) that is identical to economy I except that economy II possesses a social security system. In particular, at each date $t \geq 1$, the government taxes $\tau > 0$ units of the time t consumption good away from each young person and at the same time gives τ units of the time t consumption good to each old person then alive.

b. Does economy II possess an equilibrium with valued fiat currency? Describe the restrictions on the parameter τ , if any, that are needed to ensure the existence of such an equilibrium.

c. If an equilibrium with valued fiat currency exists, is it unique?

d. Consider the *stationary* equilibrium with valued fiat currency. Is it unique? Describe how the value of currency or price level would vary across economies with differences in the size of the social security system, as measured by τ .

Solution:

(a) An *equilibrium with valued currency* is a positive finite $\{p(t)\}$ such that, with $R_t = p(t)/p(t + 1)$, each young agent saves $s(R_t)$ and $Ns(R_t) = H/p(t)$. With endowment

$(y, 0)$, (T1) gives $s = y/2$ for every R_t . So $H/p(t) = Ny/2$ and

$$p(t) = \frac{2H}{Ny} \quad \forall t, \quad c_t(t) = c_t(t+1) = \frac{y}{2}, \quad c_0(1) = \frac{y}{2}.$$

It exists because $y > 0$. It is unique because saving is interest-inelastic: the clearing condition pins $p(t)$ date by date, with no forward-looking term.

(b) In economy II the young's net endowment is $(y - \tau, \tau)$, so $s(R_t) = \frac{1}{2}(y - \tau - \frac{\tau}{R_t})$. With $m_t = H/(Np(t))$ and $R_t = m_{t+1}/m_t$, clearing reads $2m_t = y - \tau - \tau m_t/m_{t+1}$. A stationary solution has $R = 1$ and $m = \frac{1}{2}(y - 2\tau)$, which is positive iff $\tau < y/2$. If $\tau \geq y/2$, no equilibrium exists. The autarkic MRS is $\theta_{\text{aut}} = c_2/c_1 = \tau/(y - \tau)$, and the section 9.7 existence condition $\theta_{\text{aut}} < n = 1$ is the same inequality. So for $\tau < y/2$ a monetary equilibrium exists, and for $\tau \geq y/2$ none does.

(c) No. Multiply clearing by $2p(t)/\tau$: $\frac{2H}{N\tau} = \frac{y-\tau}{\tau}p(t) - p(t+1)$, so

$$p(t+1) = \frac{y-\tau}{\tau}p(t) - \frac{2H}{N\tau}, \quad p(t) = p^* + c\left(\frac{y-\tau}{\tau}\right)^t, \quad c \geq 0, \quad p^* = \frac{2H}{N(y-2\tau)}.$$

Since $(y - \tau)/\tau > 1$, every $c > 0$ is an equilibrium in which the value of currency goes to 0. Negative c gives negative prices. There is a continuum, as in (9.4.8). Economy I is the limit $\tau = 0$, where the forward term vanishes.

(d) The stationary one ($c = 0$) is unique: $p^* = 2H/[N(y - 2\tau)]$ and $1/p^* = N(y - 2\tau)/(2H)$. The value of currency **falls** and the price level **rises** with τ . As $\tau \uparrow y/2$, $p^* \rightarrow \infty$. The allocation is *independent* of τ : $c_y = y - \tau - m = y/2$ and $c_o = \tau + m = y/2$. Social security and fiat money are two ways to make the same intergenerational transfer, and each unit of transfer by the government crowds out one unit of real balances.

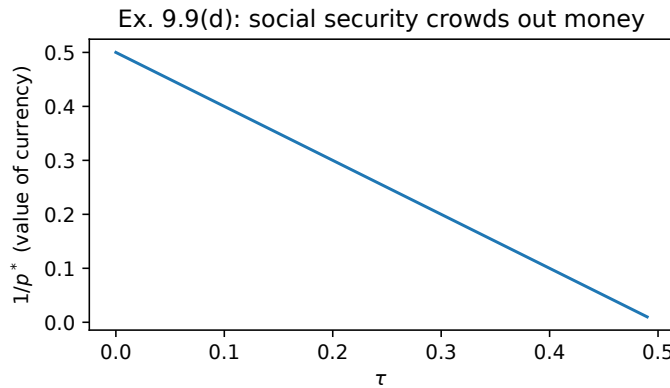


Figure 2: Exercise 9.9(d). Value of currency in the stationary equilibrium against τ ($y = N = H = 1$).

Numerical exercise — see the companion code

$y = N = H = 1$, $\tau \in \{0, .1, .3, .45\}$: stationary $m = s(1)$ (closed form and numerical), and the allocation is $(y/2, y/2)$. The closed-form continuum solves the recursion for $c \in \{0, .5\}$.

Exercise 9.10 Seignorage

Question

pp. 373–375. Consider an economy consisting of overlapping generations of two-period-lived agents. At each date $t \geq 1$, there are born N_1 “lenders” who are endowed with $\alpha > 0$ units of the single consumption good when they are young and zero units when they are old. At each date $t \geq 1$, there are also born N_2 “borrowers” who are endowed with zero units of the consumption good when they are young and $\beta > 0$ units when they are old. The good is nonstorable, and N_1 and N_2 are constant through time. The economy starts at time 1, at which time there are N old people who are in the aggregate endowed with $H(0)$ units of unbacked, intrinsically worthless pieces of paper called dollars. Assume that α, β, N_1 , and N_2 are such that

$$\frac{N_2\beta}{N_1\alpha} < 1.$$

Assume that everyone has preferences

$$u[c_t^h(t), c_t^h(t+1)] = \ln c_t^h(t) + \ln c_t^h(t+1),$$

where $c_t^h(s)$ is consumption of time s good of agent h born at time t .

a. Compute the equilibrium interest rate on consumption loans in the equilibrium without valued currency.

b. Construct a *brief* argument to establish whether or not the equilibrium without valued currency is Pareto optimal.

The economy also contains a government that purchases and destroys G_t units of the good in period t , $t \geq 1$. The government finances its purchases entirely by currency creation. That is, at time t ,

$$G_t = \frac{H(t) - H(t-1)}{p(t)},$$

where $[H(t) - H(t-1)]$ is the additional dollars printed by the government at t and $p(t)$ is the price level at t . The government is assumed to increase $H(t)$ according to

$$H(t) = zH(t-1), \quad z \geq 1,$$

where z is a constant for all time $t \geq 1$.

At time t , old people who carried $H(t-1)$ dollars between $(t-1)$ and t offer these $H(t-1)$ dollars in exchange for time t goods. Also at t the government offers $H(t) - H(t-1)$ dollars for goods, so that $H(t)$ is the total supply of dollars at time t , to be carried over by the young into time $(t+1)$.

c. Assume that $1/z > N_2\beta/N_1\alpha$. Show that under this assumption there exists a continuum of equilibria with valued currency.

d. Display the unique stationary equilibrium with valued currency in the form of a “quantity theory” equation. Compute the equilibrium rate of return on currency and consumption loans.

e. Argue that if $1/z < N_2\beta/N_1\alpha$, then there exists no valued-currency equilibrium. Interpret this result. (*Hint:* Look at the rate of return on consumption loans in the equilibrium without valued currency.)

f. Find the value of z that *maximizes* the government’s G_t in a stationary equilibrium. Compare this with the largest value of z that is compatible with the existence of a valued currency equilibrium.

Solution: Let $A \equiv N_1\alpha$ and $B \equiv N_2\beta z$. Aggregate saving at gross return R is $S(R) = \frac{1}{2}(N_1\alpha - N_2\beta/R)$ (section 9.8).

(a) $S(R) = 0$ gives $\boxed{1 + r = N_2\beta/(N_1\alpha) < 1}$.

(b) **Not optimal.** By Balasko–Shell (9.7.18), the constant interest factor $\rho = N_2\beta/(N_1\alpha) < 1$ gives $\sum_t \rho^t < \infty$. Directly, take $\varepsilon > 0$ from each young lender and hand it to the old lenders of the previous generation (to the initial old at $t = 1$: $N_1\varepsilon/N$ each). Borrowers keep their allocation. Feasibility holds at every date. A lender's utility changes by $\ln(\frac{\alpha}{2} - \varepsilon) + \ln(\frac{\rho\alpha}{2} + \varepsilon) - \ln \frac{\alpha}{2} - \ln \frac{\rho\alpha}{2} \approx \varepsilon \left(\frac{2}{\rho\alpha} - \frac{2}{\alpha} \right) > 0$ for small ε , since $\rho < 1$. The initial old gain too. This is a Pareto improvement.

(c) Let $m_t = H(t)/p(t)$ be aggregate real balances. The currency return is $R_t = \frac{p(t)}{p(t+1)} = \frac{m_{t+1}}{m_t} \frac{H(t)}{H(t+1)} = \frac{m_{t+1}}{zm_t}$, and loans pay the same. The young hold all currency: $S(R_t) = m_t$, that is $m_t = \frac{1}{2} \left(A - \frac{N_2\beta zm_t}{m_{t+1}} \right)$. Solve for m_{t+1} and invert:

$$m_{t+1} = \frac{Bm_t}{A - 2m_t} \implies x_{t+1} = \frac{A}{B}x_t - \frac{2}{B}, \quad x_t \equiv \frac{1}{m_t}.$$

This is linear, with fixed point $x^* = 2/(A - B)$, positive iff $A > B$, which is $1/z > N_2\beta/(N_1\alpha)$. Then $x_t - x^* = (A/B)^{t-1}(x_1 - x^*)$ with $A/B > 1$. Every $x_1 \geq x^*$, that is $m_1 \in (0, m^*]$ with $m^* = (A - B)/2$, keeps $x_t > 0$ and is an equilibrium. $x_1 < x^*$ drives x_t negative. This gives a **continuum** indexed by $p(1) = zH(0)/m_1 \geq zH(0)/m^*$, all but one with $m_t \rightarrow 0$. (The old's $H(0)$ plus the new issue $H(1) - H(0)$ make up $H(1)$; at $t = 1$ the condition $S(R_1) = m_1$ holds as written.)

(d) Stationary: $m_t = m^* = \frac{1}{2}(N_1\alpha - N_2\beta z)$, so

$$\boxed{p(t) = \frac{2}{N_1\alpha - N_2\beta z} H(t)}, \quad R = \frac{1}{z} \text{ on currency and on loans}, \quad G = \left(1 - \frac{1}{z}\right)m^*.$$

The price level grows at the money growth rate z .

(e) If $A < B$, then $x^* < 0$ and $A/B < 1$. The linear map sends every $x_1 > 0$ toward $x^* < 0$, so $x_t < 0$ in finite time: m_t explodes past $A/2$. If $A = B$, $x_{t+1} = x_t - 2/B$, which also turns negative. **No valued-currency equilibrium exists.** *Interpretation.* In the long run currency returns at most $1/z$. If $1/z$ is below the nonmonetary loan rate $N_2\beta/(N_1\alpha)$, borrowers' demand at that rate exceeds lenders' supply ($S(1/z) < 0$): nobody is left to hold money. The inflation tax is too high relative to the private market's interest rate.

(f) $G(z) = \frac{1}{2} \left(1 - \frac{1}{z}\right)(N_1\alpha - N_2\beta z) = \frac{1}{2} \left[N_1\alpha + N_2\beta - N_2\beta z - \frac{N_1\alpha}{z} \right]$. So $G'(z) = \frac{1}{2} \left[-N_2\beta + \frac{N_1\alpha}{z^2} \right] = 0$ and $G'' < 0$:

$$\boxed{z^* = \sqrt{\frac{N_1\alpha}{N_2\beta}}}, \quad G(z^*) = \frac{1}{2} \left(\sqrt{N_1\alpha} - \sqrt{N_2\beta} \right)^2.$$

The largest admissible z is $\bar{z} = N_1\alpha/(N_2\beta)$ (the supremum; at $\bar{z}$, $G = 0$). Since $\bar{z} > 1$, $z^* = \sqrt{\bar{z}} \in (1, \bar{z})$. The revenue-maximising money growth rate is the square root of the maximal one. Rates in $(z^*, \bar{z})$ are on the wrong side of the Laffer curve.

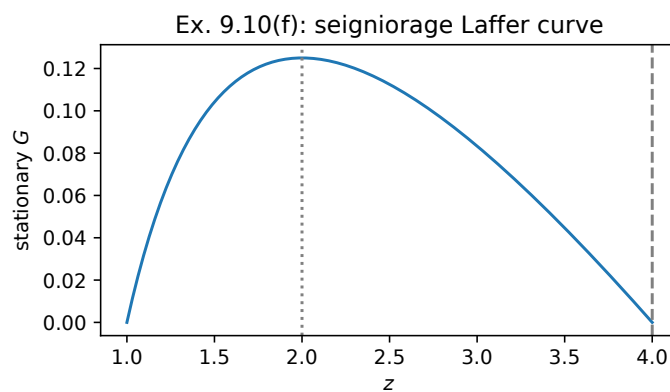


Figure 3: Exercise 9.10(f). Stationary seigniorage $G(z)$ for $N_1 = N_2 = \alpha = 1$, $\beta = .25$. Dotted: $z^* = 2$; dashed: $\bar{z} = 4$.

Numerical exercise — see the companion code

$N_1 = N_2 = \alpha = 1$, $\beta = .25$, $z = 2$: m^* is a fixed point, $m_1 = .5m^*$ gives $m_t \rightarrow 0$, $1/m_t$ satisfies the linear recursion exactly, and $m_1 = 1.01m^*$ explodes. With $z = 1.2\bar{z}$, all $m_1 \in \{10^{-4}, .01, .1, .3\}$ fail. A grid maximiser of G gives $z^* = 2$ and $G^* = .125$.

Exercise 9.11 Unpleasant monetarist arithmetic

Question

pp. 375–376. Consider an economy in which the aggregate demand for government currency for $t \geq 1$ is given by $[M(t)p(t)]^d = g[R_1(t)]$, where $R_1(t)$ is the gross rate of return on currency between t and $(t+1)$, $M(t)$ is the stock of currency at t , and $p(t)$ is the value of currency in terms of goods at t (the reciprocal of the price level). The function $g(R)$ satisfies

$$(1) \quad g(R)(1 - R) = h(R) > 0 \quad \text{for } R \in (\underline{R}, 1),$$

where $h(R) \leq 0$ for $R < \underline{R}$, $R \geq 1$, $\underline{R} > 0$ and where $h'(R) < 0$ for $R > R_m$, $h'(R) > 0$ for $R < R_m$, $h(R_m) > D$, where D is a positive number to be defined shortly. The government faces an infinitely elastic demand for its interest-bearing bonds at a constant-over-time gross rate of return $R_2 > 1$. The government finances a budget deficit D , defined as government purchases minus explicit taxes, that is constant over time. The government's budget constraint is

$$(2) \quad D = p(t)[M(t) - M(t-1)] + B(t) - B(t-1)R_2, \quad t \geq 1,$$

subject to $B(0) = 0$, $M(0) > 0$. In equilibrium,

$$(3) \quad M(t)p(t) = g[R_1(t)].$$

The government is free to choose paths of $M(t)$ and $B(t)$, subject to equations (2) and (3).

- a. Prove that, for $B(t) = 0$, for all $t > 0$, there exist two stationary equilibria for this model.
- b. Show that there exist values of $B > 0$, such that there exist stationary equilibria with $B(t) = B$, $M(t)p(t) = Mp$.
- c. Prove a version of the following proposition: among stationary equilibria, the lower the value of B , the lower the stationary rate of inflation consistent with equilibrium. (You will have to make an assumption about Laffer curve effects to obtain such a proposition.)

This problem displays some of the ideas used by Sargent and Wallace (1981). They argue that, under assumptions like those leading to the proposition stated in part c, the “looser” money is today [that is, the higher $M(1)$ and the lower $B(1)$], the lower the stationary rate of inflation.

Solution: Here $p(t)$ is the *value* of currency, so $R_1(t) = p(t+1)/p(t)$, and gross inflation is $1/R_1$. *Reading.* The hypotheses on h presume h is continuous (it is differentiable), and “ $h(R_m) > D$ ” is a separate clause (a comma is missing in the statement).

(a) In a stationary equilibrium $R_1(t) = R$, so $M(t)p(t) = g(R)$ and $p(t)M(t-1) =$

$\frac{p(t)}{p(t-1)} p(t-1)M(t-1) = Rg(R)$. With $B \equiv 0$, (2) becomes

$$D = g(R) - Rg(R) = g(R)(1 - R) = h(R).$$

h is continuous and $h(\underline{R}) \leq 0 < D < h(R_m)$. By the intermediate value theorem there is a root in $(\underline{R}, R_m)$, and it is unique there because $h' > 0$. Likewise $h(1) \leq 0 < D < h(R_m)$ gives a unique root in $(R_m, 1)$ because $h' < 0$ there. So there are **exactly two** stationary equilibria, $\underline{R} < R^{\text{bad}} < R_m < R^{\text{good}} < 1$. At $t = 1$, (2) reads $D = g(R) - p(1)M(0)$, which gives $p(1) = Rg(R)/M(0) > 0$. Then $M(t) = g(R)/[p(1)R^{t-1}]$ grows at $1/R$.

(b) With $B(t) = B$ for $t \geq 1$, (2) for $t \geq 2$ is $D = h(R) + B - R_2B$, that is

$$h(R) = D + (R_2 - 1)B.$$

The right side exceeds D because $R_2 > 1$. By the argument of (a), two stationary roots exist whenever $D + (R_2 - 1)B < h(R_m)$, i.e. for all $0 < B < \bar{B} \equiv \frac{h(R_m) - D}{R_2 - 1}$, which

is a nonempty interval because $h(R_m) > D$. At $B = \bar{B}$ the two roots merge at R_m . At $t = 1$, $D = g(R) - p(1)M(0) + B$, so $p(1)M(0) = g(R) + B - D$.

(c) Differentiate $h(R) = D + (R_2 - 1)B$: $h'(R)dR = (R_2 - 1)dB$, so $\frac{dR}{dB} = \frac{R_2 - 1}{h'(R)}$. *Assumption:* the economy is on the good side of the Laffer curve, $R \in (R_m, 1)$, where $h' < 0$. This is the equilibrium selected by the adaptive and learning dynamics discussed in section 9.5.1. Then $dR/dB < 0$, and gross inflation $1/R$ rises with B . **The lower B , the higher R and the lower stationary inflation.** On the bad side ($h' > 0$) the sign flips. Interpretation: selling bonds today ($B(1)$ up, $M(1)$ down, i.e. “tighter” money now) commits the government to interest payments $(R_2 - 1)B$ forever. These must be covered by more seigniorage $h(R)$, which on the good side requires more inflation.

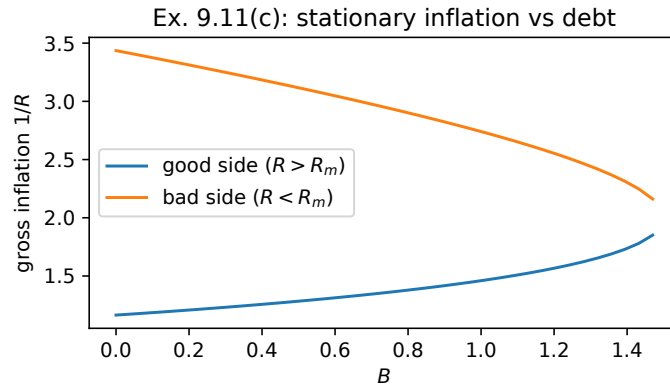


Figure 4: Exercise 9.11(c). Stationary gross inflation $1/R$ against B on both sides of the Laffer curve. Example: $g(R) = (1 - .25/R)/2$, $D = .05$, $R_2 = 1.05$.

Numerical exercise — see the companion code

Example demand (chosen): $g(R) = \frac{1}{2}(w_1 - w_2/R)$, the log-OLG saving function with $w_1 = 1$, $w_2 = .25$. Then h has $\underline{R} = .25$, $R_m = .5$ (numerical maximiser), $h(R_m) = .125$; $D = .05$, $R_2 = 1.05$, $\bar{B} = 1.5$. For 40 values of $B \in [0, .98\bar{B}]$, `brentq` roots equal the quadratic $w_1 R^2 - (w_1 + w_2 - 2[D + (R_2 - 1)B])R + w_2 = 0$. R^{good} falls from .8589 to .5403 and R^{bad} rises. $p(1)M(0) > 0$ throughout.

Exercise 9.12 Grandmont–Hall

Question

pp. 376–377. Consider a nonstochastic, one-good overlapping generations model consisting of two-period-lived young people born in each $t \geq 1$ and an initial group of old people at $t = 1$ who are endowed with $H(0) > 0$ units of unbacked currency at the beginning of period 1. The one good in the model is not storable. Let the aggregate first-period saving function of the young be time-invariant and be denoted $f[1 + r(t)]$ where $[1 + r(t)]$ is the gross rate of return on consumption loans between t and $(t + 1)$. The saving function is assumed to satisfy $f(0) = -\infty$, $f'(1 + r) > 0$, $f(1) > 0$.

Let the government pay interest on currency, starting in period 2 (to holders of currency between periods 1 and 2). The government pays interest on currency at a nominal rate of $[1 + r(t)]p(t + 1)/\bar{p}$, where $[1 + r(t)]$ is the real gross rate of return on consumption loans, $p(t)$ is the price level at t , and $\bar{p}$ is a target price level chosen to satisfy

$$\bar{p} = H(0)/f(1).$$

The government finances its interest payments by printing new money, so that the government's budget constraint is

$$H(t + 1) - H(t) = \left\{ [1 + r(t)] \frac{p(t + 1)}{\bar{p}} - 1 \right\} H(t), \quad t \geq 1,$$

given $H(1) = H(0) > 0$. The gross rate of return on consumption loans in this economy is $1 + r(t)$. In equilibrium, $[1 + r(t)]$ must be at least as great as the real rate of return on currency

$$1 + r(t) \geq [1 + r(t)]p(t)/\bar{p} = [1 + r(t)] \frac{p(t + 1)}{\bar{p}} \frac{p(t)}{p(t + 1)}$$

with equality if currency is valued,

$$1 + r(t) = [1 + r(t)]p(t)/\bar{p}, \quad 0 < p(t) < \infty.$$

The loan market-clearing condition in this economy is

$$f[1 + r(t)] = H(t)/p(t).$$

- a. Define an equilibrium.
- b. Prove that there exists a unique monetary equilibrium in this economy and compute it.

Solution:

(a) An *equilibrium* is a sequence $\{1 + r(t), p(t), H(t)\}_{t \geq 1}$ with $H(1) = H(0)$ that satisfies

four conditions. (i) The budget constraint $H(t+1) = [1 + r(t)] \frac{p(t+1)}{\bar{p}} H(t)$. (ii) The arbitrage condition $1 + r(t) \geq [1 + r(t)]p(t)/\bar{p}$, with equality when currency is valued. (iii) Loan-market clearing $f[1 + r(t)] = H(t)/p(t)$. (iv) The young's saving is optimal, which is embodied in f . It is *monetary* if $0 < p(t) < \infty$ for all t .

(b) Price level. In a monetary equilibrium the arbitrage condition holds with equality: $1 + r(t) = [1 + r(t)]p(t)/\bar{p}$. Since $1 + r(t) > 0$, this gives $p(t) = \bar{p}$ for all t . *Interest rate.* At $t = 1$, clearing gives $f[1 + r(1)] = H(1)/\bar{p} = H(0)/\bar{p} = f(1)$. Since f is strictly increasing, $1 + r(1) = 1$. *Induction.* Suppose $1 + r(t) = 1$ and $H(t) = H(0)$. Then (i) gives $H(t+1) = 1 \cdot \frac{\bar{p}}{\bar{p}} H(t) = H(0)$, and clearing at $t+1$ gives $f[1 + r(t+1)] = f(1)$, so $1 + r(t+1) = 1$. Hence

$$p(t) = \bar{p} = \frac{H(0)}{f(1)}, \quad 1 + r(t) = 1, \quad H(t) = H(0) \quad \forall t \geq 1$$

is the unique monetary equilibrium. It exists because $f(1) > 0$ makes $\bar{p}$ positive and finite. *Why uniqueness.* Paying interest indexed to the price gap removes the continuum of 9.1(e). Any $p(t) \neq \bar{p}$ would give currency a real return different from loans. Given $p = \bar{p}$, the system $f[1 + r(t+1)] = [1 + r(t)]f[1 + r(t)]$ is *unstable* at $1 + r = 1$ (slope $1 + f(1)/f'(1) > 1$). Paths starting at $1 + r(1) \neq 1$ either explode or drift to autarky, and the initial condition $H(1) = H(0)$ rules them all out. The condition $f(0) = -\infty$ guarantees that the nonmonetary loan market has a positive clearing rate. It is not needed for the monetary equilibrium.

Numerical exercise — see the companion code

$f(R) = \frac{1}{2}(1 - .5/R)$ (so $f(1) = .25$), $H(0) = 1$: $\bar{p} = 4$. `brentq` gives $1 + r(1) = 1$, and the recursion maps 1 to 1. Starting at 1 ± 10^{-6} , the deviation doubles per period (slope $1 + f(1)/f'(1) = 2$). The upper path becomes infeasible in finite time. The lower one converges to the autarkic $w_2/w_1 = .5$ but violates $f[1 + r(1)] = H(0)/\bar{p}$.

Exercise 9.13 Bryant–Keynes–Wallace

Question

pp. 377–378. Consider an economy consisting of overlapping generations of two-period-lived agents. There is a constant population of N young agents born at each date $t \geq 1$. There is a single consumption good that is not storable. Each agent born in $t \geq 1$ is endowed with w_1 units of the consumption good when young and with w_2 units when old, where $0 < w_2 < w_1$. Each agent born at $t \geq 1$ has identical preferences $\ln c_t^h(t) + \ln c_t^h(t+1)$, where $c_t^h(s)$ is time s consumption of agent h born at time t . In addition, at time 1, there are alive N old people who are endowed with $H(0)$ units of unbacked paper currency and who want to maximize their consumption of the time 1 good.

A government attempts to finance a constant level of government purchases $G(t) = G > 0$ for $t \geq 1$ by printing new base money. The government's budget constraint is

$$G = [H(t) - H(t-1)]/p(t),$$

where $p(t)$ is the price level at t , and $H(t)$ is the stock of currency carried over from t to $(t+1)$ by agents born in t . Let $g = G/N$ be government purchases per young person. Assume that purchases $G(t)$ yield no utility to private agents.

- a. Define a stationary equilibrium with valued fiat currency.
- b. Prove that, for g sufficiently small, there exists a stationary equilibrium with valued fiat currency.
- c. Prove that, in general, if there exists one stationary equilibrium with valued fiat currency, with rate of return on currency $1 + r(t) = 1 + r_1$, then there exists at least one other stationary equilibrium with valued currency with $1 + r(t) = 1 + r_2 \neq 1 + r_1$.
- d. Tell whether the equilibria described in parts b and c are Pareto optimal, among allocations among private agents of what is left after the government takes $G(t) = G$ each period. (A proof is not required here: an informal argument will suffice.)

Now let the government institute a forced saving program of the following form. At time 1, the government redeems the outstanding stock of currency $H(0)$, exchanging it for government bonds. For $t \geq 1$, the government offers each young consumer the option of saving at least F worth of time t goods in the form of bonds bearing a constant rate of return $(1 + r_2)$. A legal prohibition against private intermediation is instituted that prevents two or more private agents from sharing one of these bonds. The government's budget constraint for $t \geq 2$ is

$$G/N = B(t) - B(t-1)(1 + r_2),$$

where $B(t) \geq F$. Here $B(t)$ is the saving of a young agent at t . At time $t = 1$, the government's budget constraint is

$$G/N = B(1) - \frac{H(0)}{Np(1)},$$

where $p(1)$ is the price level at which the initial currency stock is redeemed at $t = 1$. The government sets F and r_2 .

Consider stationary equilibria with $B(t) = B$ for $t \geq 1$ and r_2 and F constant.

- e. Prove that if g is small enough for an equilibrium of the type described in part a to exist, then a stationary equilibrium with forced saving exists. (Either a graphical argument or an algebraic argument is sufficient.)
- f. Given g , find the values of F and r_2 that maximize the utility of a representative

Solution: Let $R = 1 + r$ be the return on currency, $s(R) = \frac{1}{2}(w_1 - w_2/R)$ from (T1), and $m_t = H(t)/(Np(t))$.

(a) A *stationary equilibrium with valued currency* is a constant $R \in (0, \infty)$ and constant real balances $m > 0$ per young person, with $p(t) = p(1)R^{1-t}$ and $H(t) = Nmp(t)$, such that three conditions hold. (i) The young optimise: $m = s(R)$. (ii) The budget holds: $g = \frac{H(t)-H(t-1)}{Np(t)} = m - m \frac{p(t-1)}{p(t)} = m(1-R)$ for $t \geq 2$. (iii) At $t = 1$, $Nm = H(0)/p(1) + G$, which fixes $p(1)$. Combining (i) and (ii):

$$g = s(R)(1 - R) \equiv h(R) = \frac{1}{2} \left(w_1 - \frac{w_2}{R} \right) (1 - R),$$

which is (9.5.5a) with $d = g$.

(b) h is continuous on $[w_2/w_1, 1]$, zero at both ends and positive inside. $h'(R) = \frac{1}{2} \left(-w_1 + \frac{w_2}{R^2} \right) = 0$ at $R_m = \sqrt{w_2/w_1}$, where $h_{\max} = \frac{1}{2}(\sqrt{w_1} - \sqrt{w_2})^2$. For $0 < g \leq h_{\max}$, a root $R \in (w_2/w_1, 1)$ exists by the intermediate value theorem. It has $s(R) > 0$ and gives $p(1) = H(0)/(N[s(R) - g]) = H(0)/(NRs(R)) > 0$. So for g small enough ($g \leq h_{\max}$) a stationary monetary equilibrium exists.

(c) Multiply $h(R) = g$ by $2R$: $w_1R - w_1R^2 - w_2 + w_2R = 2gR$, that is

$$w_1R^2 - (w_1 + w_2 - 2g)R + w_2 = 0, \quad R_1R_2 = \frac{w_2}{w_1}.$$

If $R_1 = 1 + r_1$ is a root, then $R_2 = w_2/(w_1R_1)$ is the other one. It lies in $(w_2/w_1, 1)$ because R_1 does, and it is also an equilibrium. $R_2 \neq R_1$ unless $R_1 = \sqrt{w_2/w_1}$, i.e. unless $g = h_{\max}$ exactly, the tangency case. That is why the statement says “in general”.

(d) **Neither is Pareto optimal.** Stationary feasibility after G is $c_y + c_o = w_1 + w_2 - g$. The golden-rule allocation $c_y = c_o = \bar{c} \equiv (w_1 + w_2 - g)/2$, with $\bar{c}$ also given to the initial old, gives every generation the maximal stationary utility $2 \ln \bar{c}$. The equilibria have $MRS = R < 1$, so they are off that point: $v(R) < 2 \ln \bar{c}$. The initial old get $w_2 + s(R) - g$, and $\bar{c} - [w_2 + s(R) - g] = \frac{1}{2} \left[\frac{w_2(1-R)}{R} + g \right] > 0$. So the golden rule Pareto dominates both. (Balasko–Shell: $\sum R^t < \infty$.) Between the two, the high-return equilibrium $R_{hi} > R_m$ Pareto dominates the low-return one. By (T2), savers' v rises with R , and the initial old's $w_2 + s(R) - g$ rises with R .

(e) Take either stationary monetary equilibrium $\bar{R}$ from (b) and set $F = s(\bar{R})$, $1 + r_2 = \bar{R}$. At return $\bar{R}$, the young's unconstrained optimum is exactly $s(\bar{R}) = F$, so they buy $B = F$ (which satisfies $B \geq F$). The budget for $t \geq 2$ is $B - B(1 + r_2) = s(\bar{R})(1 - \bar{R}) = h(\bar{R}) = g$. At $t = 1$, $p(1) = H(0)/(N[B - g])$, and $B - g = \bar{R}s(\bar{R}) > 0$. So a stationary forced-saving equilibrium exists and replicates the monetary allocation. Graphically, it is the same point as in Figure 9.2.4, with bonds in place of currency.

(f) Stationarity gives $g = B - B(1 + r_2)$, so $(1 + r_2)B = B - g$. The agent consumes $c_y = w_1 - B$ and $c_o = w_2 + (1 + r_2)B = w_2 + B - g$. The government can implement any B by setting $F = B$ and $1 + r_2 = 1 - g/B$. Maximise $\ln(w_1 - B) + \ln(w_2 + B - g)$:

the first-order condition $\frac{1}{w_1 - B} = \frac{1}{w_2 + B - g}$ gives

$$F = B^* = \frac{w_1 - w_2 + g}{2}, \quad 1 + r_2 = 1 - \frac{g}{B^*} = \frac{w_1 - w_2 - g}{w_1 - w_2 + g}, \quad c_y = c_o = \bar{c}.$$

Implementability. At $1 + r_2 < 1$ the agent's desired saving $s(1 + r_2)$ is below F . The minimum binds and the agent buys exactly F , provided this beats not buying at all: $2 \ln \bar{c} \geq \ln w_1 + \ln w_2$. That holds iff $g \leq (\sqrt{w_1} - \sqrt{w_2})^2 = 2h_{\max}$, which is implied by $g \leq h_{\max}$. The indivisibility of bonds, plus the ban on sharing them, is what stops agents from undoing the minimum. That is the role of the legal restriction.

(g) Yes. The allocation gives $\bar{c}$ to both ages, and the initial old get $w_2 + B^* - g = \bar{c}$ as well. Its MRS equals $1 = n$: an interest factor of 1 makes $\sum_t \prod_s (1 + r_s) = \infty$, so it is optimal by Balasko–Shell (constant population). Informally, any reallocation that helps one generation must take from the next. The chain argument of section 9.7 ($\theta = n$) shows the deficit then grows without bound, which is infeasible. This is the stationary counterpart of 9.4–9.5, where the optimal equilibrium has return n . Forced saving at a negative real rate collects g as a lump sum on savers without distorting the margin, unlike the inflation tax.

Numerical exercise — see the companion code

$w_1 = 1$, $w_2 = .25$, $g = .05$: $R_{\text{lo}} = .29105$ and $R_{\text{hi}} = .85895$, with product $.25$. The high root dominates for every generation and the initial old. The golden rule $\bar{c} = .6$ dominates both. $F = s(\bar{R})$, $1 + r_2 = \bar{R}$ satisfy the budget. A numerical maximiser gives $B^* = .4 = F$ and $1 + r_2 = .875$. At $.875$ the unconstrained saving is below F , and $2 \ln \bar{c} > \ln w_1 + \ln w_2$.

Companion code

Every numerical result above is reproduced by the notebook `ls_ch09.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Section	What it checks
9.1	CRRA saving closed form vs numerical optimisation; $s(1) = (w_1 - w_2)/2$; forward map $m_t \rightarrow 0$, $R_t \rightarrow (w_2/w_1)^\gamma$; figure.
9.2	Loan-market root; price-path closed form vs recursion; money-market clearing; lenders vs borrowers welfare.
9.3	Quadratic vs brentq tree price ($\gamma = 1, 2$); instability of $a_1 \neq \bar{a}$; unbounded tree value.
9.4–9.5	$R = n$; closed-form p_t vs recursion; limits of R_t ; Pareto ranking in c ; nonexistence for $nw_1 \leq w_2$.
9.6	Kareken–Wallace continuum in e ; constant-split e ; nominal-anchor prices in (d).
9.7–9.8	Loan-market roots; $x \mapsto x/(1 - x)$ exit; $q = 8/y$; feasibility; welfare; inside-money clearing.
9.9–9.10	Social-security continuum and crowding out (figure); linear $1/m$ dynamics; Laffer maximum (figure).
9.11	Two stationary roots vs quadratic; comparative statics in B (figure).
9.12	$p = \bar{p}$, $1 + r = 1$; instability of the recursion.
9.13	Root product; Pareto comparisons; forced-saving replication and optimum.

Figures are saved to `fig/ls09_*.pdf` with TrueType (not Type 3) fonts.